# APEX

## Adaptive Performance & Experience Profiling System

*Production-Grade System Specification*

**Failsafe Architecture Edition**

MSc Advanced Data Science & AI | University of Liverpool | 2026

**Implementation Guide for Fable 5 AI — End-to-End Build**

# 1. Executive Summary

APEX is a production-grade F1 driver behavioural profiling platform. It solves three interconnected problems that every F1 team faces every race weekend. It is built to be unbreakable: every pipeline stage has defined failure modes, fallback behaviour, and explicit output contracts. No component fails silently.

## 1.1 The Three Problems APEX Solves

Problem A — The Driver-Engineer Communication Gap: teams have no objective language to translate driver feel into setup parameters. A driver says the car is nervous on entry. The engineer guesses. APEX quantifies that statement.

Problem B — New Driver Onboarding: when a driver changes teams, it takes 6–12 months to configure the car around their style. APEX compresses this by projecting any driver's telemetry into a universal style space and generating a setup starting point immediately.

Problem D — Tyre Degradation Detection: degradation manifests in driving behaviour before lap time. APEX monitors fingerprint deviation as a leading indicator, alerting engineers before the lap time signal appears.

## 1.2 What Makes This System Production-Grade

- Every data pipeline stage has explicit acceptance criteria. Data that fails the contract is rejected with a specific error code, not silently dropped.
- Every ML model output is gated by a confidence threshold. Below the threshold, the system returns an explicit uncertainty state — not a wrong answer.
- Every module has a documented fallback mode. The system degrades gracefully, never crashes silently.
- Every training run is seeded, logged, and reproducible from the public GitHub repository.
- Every evaluation result includes failure cases, not just success metrics. The dissertation is honest.

## 1.3 Scope Boundary

APEX covers driver fingerprinting, style space projection, car personalisation, tyre degradation detection, and historical race replay monitoring. Synthetic 2026 data generation is explicitly out of scope. Real-time live race integration is out of scope; historical replay achieves equivalent demo value with zero infrastructure risk. Integration with the pit stop and ERS partner system is a post-submission milestone with a defined API contract.

## 1.4 Novel Academic Contributions

- A unified behavioural telemetry framework that quantifies driver-car compatibility as a continuous score, not a binary classification.

- Fingerprint deviation as a leading indicator for tyre degradation, validated against three seasons of official F1 race data.
- Cross-series style transfer: projecting unknown drivers into the same style space as established F1 drivers without retraining.
- Comprehensive failsafe architecture for ML-driven sports telemetry systems — a methodological contribution independent of the F1 domain.

# 2. Problem Statement & Research Questions

## 2.1 Problem A — Driver-Engineer Communication Gap

Driver feedback is verbal and subjective. An engineer hearing 'the car feels loose on entry' has no objective telemetry reference. They make a setup change based on interpretation. If wrong, another 15 minutes of practice is consumed repeating the cycle. Across a weekend, this costs two to four setup iterations that could have been spent validating the race setup.

APEX solves this by maintaining a quantified fingerprint per driver that maps directly to setup parameters. Before the driver speaks, the engineer already has an objective starting point derived from the driver's own historical behaviour.

## 2.2 Problem B — New Driver Onboarding

Every driver change triggers a period of expensive trial and error. A fingerprinting system that can project any driver's telemetry — from any car, any series — into the same universal style space generates a setup starting point from the first session. The system does not need the driver to have driven this specific car before.

## 2.3 Problem D — Tyre Degradation Detection

Tyre degradation shows up in driving behaviour before lap time. A driver unconsciously widens their throttle application, softens trail braking, and reduces apex speed as grip drops. By the time these changes appear as a lap time delta, the optimal pit window may have passed. APEX monitors fingerprint deviation as a leading indicator and generates an alert 2–5 laps before the behavioural signal becomes a lap time signal.

## 2.4 Research Questions

- RQ1: Can a stable, discriminative behavioural fingerprint be extracted from F1 race telemetry that generalises to circuits not seen during training?
- RQ2: How strongly does fingerprint-to-setup compatibility correlate with stint performance consistency across drivers and configurations?
- RQ3: Does fingerprint deviation during a race stint reliably precede measurable tyre degradation in lap time, and by how many laps?
- RQ4: Is the style space representation sufficiently continuous to project unknown or feeder-series drivers into meaningful positions relative to known F1 drivers?
- RQ5: What are the principal failure modes of a behavioural telemetry profiling system, and how do explicit failsafe mechanisms affect output reliability?

> **⚠ RQ5 is the dissertation's distinguishing research question. No published system has formally characterised the failure modes of F1 telemetry ML systems. This makes APEX methodologically novel beyond its application.**

# 3. Data Architecture

## 3.1 Data Sources

| Source | Resolution | Usage | Failure Mode & Fallback |
|---|---|---|---|
| FastF1 Python Library | ~50Hz | Primary telemetry: Speed, Throttle, Brake, RPM, Gear, DRS, X/Y position | If session unavailable: log error code DATA_001, skip session. Do not substitute. Alert operator. |
| OpenF1 REST API | ~3.7Hz | Tyre compound, pit in/out flags, weather, session status | If API timeout (>10s): use Ergast API as fallback. If both fail: mark compound as UNKNOWN, exclude from compound-stratified analysis only. Core fingerprint unaffected. |
| Ergast API (via FastF1) | Per-lap | Lap times, stint numbers, tyre age. Cross-reference for compound validation. | If unavailable: derive tyre age from pit flags in FastF1 data. Flag laps with derived tyre age with DERIVED_AGE tag. |

## 3.2 Data Coverage

| | |
|---|---|
| **Circuits (10)** | Bahrain, Saudi Arabia, Canada, Monaco, Japan, Spain, Great Britain, Belgium, Italy, São Paulo |
| **Drivers (8)** | VER, HAM, ALO, LEC, RUS, NOR, SAI, PER |
| **Seasons** | 2022, 2023, 2024 |
| **Session type** | Race only. Green-flag laps only. No SC, VSC, pit, outlap, or inlap. |
| **Minimum clean laps required** | 30 per driver per circuit. If below: circuit-driver combination excluded from training. Logged as DATA_INSUFFICIENT. |
| **Expected clean lap count** | Approximately 7,000–8,400 rows in final feature matrix after filtering |
| **Circuit mix** | 4 high-speed (Bahrain, Belgium, Britain, Italy), 3 technical (Japan, Spain, Canada), 2 street (Monaco, Saudi Arabia), 1 mixed (São Paulo) |
| **Held-out test set** | Belgium and Italy withheld from all training. Used only for final generalisation evaluation. |

## 3.3 Lap Filtering Rules — Data Quality Contract

Every lap must pass all filters to enter the feature matrix. Filters are applied in order. First failure rejects the lap with a specific error code. No lap is silently dropped.

| Filter | Rule | Error Code | Action on Failure |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Session flag | Lap must occur entirely under green flag. SC, VSC, red flag laps excluded. | LAP_001 | Reject lap. Log with lap number and reason. |
| Lap type | Outlaps and inlaps excluded. Defined as lap containing a pit in or pit out event. | LAP_002 | Reject lap. Log with lap number and reason. |
| Telemetry completeness | No channel (Speed, Throttle, Brake, RPM, Gear) may have more than 10% consecutive NaN samples. | LAP_003 | Reject lap. Log channel name and dropout percentage. |
| Telemetry completeness (minor) | Gaps of 1–5 consecutive NaN samples in any channel filled via linear interpolation. | LAP_INTERP | Accept lap. Tag with INTERPOLATED flag. Log gap size. |
| Sampling rate | Effective sampling rate must be ≥20Hz after interpolation. Below this: insufficient resolution for feature extraction. | LAP_004 | Reject lap. Log computed sampling rate. |
| Speed anomaly | Lap minimum speed must be >0 km/h for >99% of samples. Stationary samples indicate telemetry error. | LAP_005 | Reject lap. Log anomaly location. |
| Lap time plausibility | Lap time must be within 115% of the session's median lap time for that driver. Extreme outliers indicate incident laps. | LAP_006 | Reject lap. Log lap time and session median. |
| Tyre age | Tyre age must be deterministic (not UNKNOWN) for Features F27, F28 only. Lap still accepted for other features. | LAP_PARTIAL | Accept lap. Mark F27, F28 as NULL for this lap. Log. |
| Minimum lap count post-filter | Driver-circuit combination must have ≥30 clean laps after all filters. Below this: exclude combination entirely. | COMBO_001 | Exclude combination from training. Log excluded combinations in data quality report. |

## 3.4 Manual Upload Pipeline — Edge Case Handling

When a user uploads a CSV file via the APEX interface, the same filter chain runs with additional pre-processing steps.

| Edge Case | Detection Method | System Response |
|---|---|---|
| Wrong column names | Check for exact column name match: Speed, Throttle, Brake, RPM, Gear required. | Return error UPLOAD_001 with list of missing columns. Do not proceed. |
| Wrong units (speed in mph) | If Speed median > 350, likely mph not km/h. Flag for user confirmation. | Return warning UPLOAD_002: 'Speed values appear to be in mph. Convert before uploading?' |
| Mixed sampling rates | Compute inter-sample time delta. If std > 20% of mean: inconsistent sampling. | Resample to 50Hz via linear interpolation. Tag all laps with RESAMPLED flag. |
| Corrupt rows (non-numeric) | Any non-numeric value in numeric columns detected on read. | Drop row. If >5% of rows corrupt: return error UPLOAD_003. If <5%: log and continue. |
| Single-lap upload | Row count insufficient for tyre age slope features (F27, F28). | Accept upload. Compute F01–F26 only. Return partial fingerprint with explicit note: 'Tyre degradation features require multiple laps.' |
| Unknown driver — no label | User selects Unknown Driver mode explicitly. | Run full pipeline without driver label. Project into style space. Return fingerprint with uncertainty bounds and nearest-neighbour matches. |
| Duplicate lap numbers | Lap number column contains repeated values. | De-duplicate by keeping first occurrence. Log duplicate count. |
| Empty file | Zero data rows after header. | Return error UPLOAD_004 immediately. Do not proceed. |

# 4. Feature Engineering

28 features are extracted per clean lap. Every feature has a defined computation method, a valid range, and an out-of-range handling rule. Features that cannot be computed for a specific lap return NULL and are excluded from that lap's vector — the lap is not rejected unless more than 6 features are NULL.

## 4.1 Feature Computation Contract

> **⚠ Every feature function must be independently unit-tested. A feature returning an out-of-range value must raise a FeatureComputationError, not silently return the raw value. The pipeline catches this error, marks the feature NULL, and logs which lap and feature triggered it.**

## 4.2 Full Feature Matrix

| ID | Name | Domain | Method | Valid Range | NULL Trigger |
|---|---|---|---|---|---|
| F01 | throttle_mean | Throttle | Mean throttle % across all samples in lap | 0–100 | All samples NaN |
| F02 | throttle_std | Throttle | Standard deviation of throttle % | 0–50 | Constant signal (std = 0 indicates sensor fault) |
| F03 | throttle_open_rate | Throttle | Mean rate of increase when throttle is rising (% per 50ms) | 0–200 | No throttle-increasing events found |
| F04 | throttle_full_fraction | Throttle | Fraction of lap samples at 100% throttle | 0–1 | None |
| F05 | throttle_lift_fraction | Throttle | Fraction of lap samples below 5% throttle | 0–1 | None |
| F06 | brake_peak_mean | Braking | Mean of top-10 brake pressure peaks per lap | 0–200 bar | Fewer than 5 braking events detected |
| F07 | brake_duration_mean | Braking | Mean duration of braking events (ms) | 50–4000ms | No braking events detected |
| F08 | brake_release_rate | Braking | Mean rate of brake pressure decrease when releasing (bar per 50ms) | 0–100 | No brake release events |
| F09 | brake_entry_speed_mean | Braking | Mean speed at point of first brake | 0–380 km/h | Corner detection failed |

| | | | | | |
|---|---|---|---|---|---|
| | | | application per corner | | |
| F10 | brake_overlap_fraction | Braking | Fraction of samples with both throttle >5% and brake >5% simultaneously | 0–0.25 | None — values >0.25 indicate sensor fault; cap and log |
| F11 | speed_min_mean | Speed | Mean of per-corner minimum speeds | 0–200 km/h | Corner detection failed |
| F12 | speed_variance | Speed | Variance of speed distribution across full lap | 0–8000 | All speed samples identical — sensor fault |
| F13 | speed_p10 | Speed | 10th percentile speed (slow corner characteristic) | 0–150 km/h | None |
| F14 | speed_p90 | Speed | 90th percentile speed (high-speed characteristic) | 100–380 km/h | None |
| F15 | gear_change_rate | Gear | Number of gear changes per kilometre | 0–15 | Lap distance cannot be computed |
| F16 | gear_hold_duration_mean | Gear | Mean time held in each gear (ms) | 100–10000ms | Fewer than 10 gear changes in lap |
| F17 | rpm_redline_fraction | Engine | Fraction of lap at >97% of session-maximum RPM | 0–0.6 | Session max RPM not deterministic |
| F18 | rpm_std | Engine | Standard deviation of RPM across lap | 0–5000 | Constant RPM — likely sensor fault |
| F19 | lateral_g_mean | Cornering | Mean lateral G-force derived from X/Y position trace and speed | 0–6G | X/Y position data unavailable |
| F20 | lateral_g_peak_mean | Cornering | Mean of top-5 lateral G peaks per lap | 0–6G | Fewer than 5 distinct lateral G peaks |
| F21 | steering_correction_rate | Cornering | Rate of steering direction reversals per second — proxy from lateral G sign changes | 0–5 | Lateral G not computable |
| F22 | throttle_speed_corr | Interaction | Pearson correlation between throttle % | -1 to 1 | None |

| | | | | | |
|---|---|---|---|---|---|
| | | | and speed across lap | | |
| F23 | brake_speed_corr | Interaction | Pearson correlation between brake pressure and speed delta (deceleration) | -1 to 1 | No braking events |
| F24 | throttle_brake_transition | Interaction | Mean time between full brake release and first throttle application (ms) | 0–3000ms | Fewer than 5 transitions in lap |
| F25 | corner_entry_balance | Interaction | Ratio: brake duration in first 50% of corner vs second 50% | 0–1 | Corner segmentation failed |
| F26 | lap_phase_throttle_delta | Phase | Throttle mean in first 50% of lap minus second 50% | -30 to 30 | None |
| F27 | tyre_age_throttle_slope | Degradation | Linear regression slope of throttle_mean vs tyre age (laps) across current stint | Unconstrained | Tyre age UNKNOWN or stint <8 laps |
| F28 | tyre_age_brake_slope | Degradation | Linear regression slope of brake_peak_mean vs tyre age across current stint | Unconstrained | Tyre age UNKNOWN or stint <8 laps |

## 4.3 Circuit-Aware Normalisation

All speed-dependent features (F09, F11, F13, F14) and lateral G features (F19, F20) are Z-score normalised per circuit. Normalisation parameters (mean, std) are computed exclusively from the training set and saved as artefacts. The same saved parameters are applied to validation, test, and new data. Under no circumstances are normalisation parameters recomputed using test or validation data.

> **⚠ Normalisation leakage is the single most common source of inflated accuracy in tabular ML systems. The normalisation parameters are computed on training circuits only, saved to /artefacts/normalisation.json, and version-controlled. Any recomputation using held-out data invalidates the evaluation.**

## 4.4 Corner Type Decomposition

Laps are segmented into three corner type categories using curvature computed from the X/Y position trace. Each produces a separate feature sub-vector. This allows the system to identify where driver differences are most pronounced.

| | |
|---|---|
| **Slow corners** | Radius < 100m. Examples: Monaco Hairpin, Monza Variante Ascari |
| **Medium corners** | Radius 100–200m. Examples: Silverstone Maggots, Spa Raidillon |
| **High-speed corners** | Radius > 200m. Examples: Suzuka 130R, Monza Parabolica |
| **Failure handling** | If curvature computation fails (X/Y data unavailable): corner decomposition is skipped. Lap-level features F01–F26 still computed. Log failure as CORNER_001. System continues. |

## 4.5 Feature Null Tolerance Policy

A lap vector with up to 6 NULL features (out of 28) is still accepted into the feature matrix. NULL features are imputed using the median value from that driver's training laps for that circuit type. If no driver-specific median is available (e.g. new circuit), the global training median is used. Imputed values are flagged with an IMPUTED tag in the database.

> **⚠ A lap vector with more than 6 NULL features is rejected entirely. Error code LAP_NULL_EXCESS. This threshold is not arbitrary: 6 NULLs represents one full feature domain failing. Beyond that, the vector is not reliable for fingerprinting.**

# 5. Machine Learning Architecture

APEX uses a three-model ML architecture. Each model has a single clearly defined responsibility. No model is asked to do more than one thing. Inter-model communication uses explicit data contracts, not shared state.

## 5.1 Model 1 — Driver Fingerprint Classifier (XGBoost)

### Purpose & Responsibility

Given a 28-feature lap vector, predict which of the 8 known drivers produced it. Return a probability distribution across all 8 drivers, not just the top prediction.

| | |
|---|---|
| **Algorithm** | XGBoost (gradient-boosted decision trees). CalibratedClassifierCV wrapper for probability calibration. |
| **Task** | 8-class classification with calibrated probability output |
| **Input** | 28-feature lap vector, circuit type encoding (3 dimensions), season (one-hot 3) |
| **Output** | Probability vector across 8 drivers + top prediction label + confidence score + SHAP values |
| **Confidence gate** | If max probability < 0.65: return LOW_CONFIDENCE state. Show top-3 candidates with probabilities. Never return a forced prediction below this threshold. |
| **Training data** | 8 drivers × 10 circuits × 3 seasons. Belgium and Italy fully withheld. |
| **Cross-validation** | Stratified 5-fold. Stratified by driver × circuit combination to prevent circuit leakage. |
| **Hyperparameter tuning** | Optuna Bayesian search. Minimum 200 trials. Objective: macro-averaged F1 on validation fold. |
| **Experiment tracking** | MLflow. Every trial logged: parameters, macro-F1, per-class F1, confusion matrix, model artefact. Best model registered as champion. |
| **Seed** | All random seeds fixed: numpy seed=42, XGBoost seed=42, Optuna sampler seed=42. Documented in config.yaml. |
| **Target metric** | Macro-averaged F1 > 0.78 on held-out circuits (Belgium, Italy) |
| **Baselines** | Majority-class classifier, Logistic Regression (same features), Random Forest (same features, default params). All logged in MLflow. |

### Model 1 Failure Modes & Responses

| Failure Mode | Detection | System Response | User-Facing Message |
|---|---|---|---|
| Max probability < 0.65 | Confidence gate check on every prediction | Return LOW_CONFIDENCE state with top-3 | 'Insufficient telemetry signal for confident identification. Top candidates shown.' |

| | | | |
|---|---|---|---|
| | | candidates and probabilities | |
| All probabilities equal (~12.5% each) | Entropy check: if Shannon entropy > 2.9 bits (near-uniform distribution) | Return AMBIGUOUS state. Likely input is not F1 telemetry. | 'Input data does not match F1 telemetry patterns. Check data source.' |
| Model file not found at startup | File existence check in startup health check | System refuses to start. Return HEALTH_FAIL. Log MODEL_NOT_FOUND. | Health endpoint returns 503 with reason. |
| Feature vector has >6 NULLs | Null count check before inference | Do not call model. Return INVALID_INPUT with null feature list. | 'Feature extraction incomplete. N features could not be computed: [list].' |
| Prediction time > 200ms | Per-request timing middleware | Log PERF_WARN. Return result with latency flag in response header. | No user-facing impact. Logged for monitoring. |

## 5.2 Model 2 — Driver Style Space (UMAP)

### Purpose & Responsibility

Project every driver's aggregate feature distribution into a continuous style space. The 5D space is the fingerprint coordinate used by all downstream modules. The 2D projection is used for visualisation. Unknown drivers can be projected in without retraining the model.

| | |
|---|---|
| **Algorithm** | UMAP (Uniform Manifold Approximation and Projection). umap-learn library. |
| **Input** | Per-driver aggregate feature distributions: mean and std of each of the 28 features, computed across all clean training laps. Shape: 8 drivers × 56 values. |
| **5D output** | Driver fingerprint vector — the coordinate used by the personalisation engine and degradation detector |
| **2D output** | Visualisation projection for the animated style space map |
| **Unknown driver projection** | New telemetry is feature-extracted, aggregated, then projected using UMAP's transform() method. The UMAP model is not retrained. This is a forward pass only. |
| **Model persistence** | UMAP model saved as /artefacts/umap_model.pkl after training. Loaded at API startup. Loaded once, shared across all requests. |
| **Stability validation** | UMAP is run 10 times with different random seeds. Procrustes alignment applied to each run. Variance in aligned driver coordinates quantifies projection stability. Reported in dissertation. |
| **Teammate proximity test** | Evaluation check: mean intra-team Euclidean distance in 5D space must be less than mean inter-team distance. Failure of this test is a meaningful research finding, not a system failure. |

### Model 2 Failure Modes & Responses

| Failure Mode | Detection | System Response |
|---|---|---|
| Unknown driver telemetry too sparse (<30 laps) | Lap count check before projection | Return SPARSE_PROJECTION warning. Compute projection but flag confidence as LOW. Show wider uncertainty ellipse in UI. |
| Unknown driver feature vector outside training distribution | Mahalanobis distance check against training distribution. Threshold: >3 standard deviations on any principal component. | Return OUT_OF_DISTRIBUTION warning. Projection computed but marked unreliable. UI shows dashed outline on style space marker. |
| UMAP model file corrupt or missing | Hash verification of model file at startup | Fall back to PCA-based projection (2-component). Log UMAP_FALLBACK. PCA model always maintained as backup artefact. |
| Transform() call fails for unknown driver | Try/except around transform() call | Fall back to nearest-neighbour in raw feature space using cosine distance. Return PROJECTION_FALLBACK state. |

## 5.3 Model 3 — Tyre Degradation Detector

### Purpose & Responsibility

Monitor fingerprint deviation within a race stint to generate a degradation alert before the signal appears in lap time. This is not a separate ML model — it is a statistical deviation detector operating on the fingerprint vectors produced by Models 1 and 2.

| | |
|---|---|
| **Method** | Cosine distance between current lap's normalised feature vector and the within-stint baseline fingerprint (computed from first 3 clean laps of stint). |
| **Rolling window** | 3-lap rolling mean of cosine distance. Prevents single-lap anomalies from triggering false alerts. |
| **Alert threshold** | 2 standard deviations above the driver's historical intra-stint variation for that circuit type. Computed from training data. Stored per driver per circuit type. |
| **Minimum stint laps before activation** | 8 laps. Below this, insufficient data for reliable baseline or rolling mean. |
| **Alert confirmation** | Deviation must exceed threshold for 2 consecutive laps before alert fires. Single-lap threshold crossing is logged but not alerted. |
| **Alert payload** | Laps since threshold crossed, SHAP decomposition of deviation (which features are driving it), pit window recommendation based on historical deg patterns, confidence of alert (LOW / MEDIUM / HIGH based on how far above threshold). |
| **F27 and F28 integration** | When stint length ≥ 8 laps, F27 and F28 (tyre age slope features) are computed on the live stint window and incorporated into the deviation score as a weighted component. |

## Degradation Detector Failure Modes

| Failure Mode | Detection | System Response |
|---|---|---|
| Stint length < 8 laps | Lap count check | Return MONITOR_INACTIVE state. Fingerprint displayed but no degradation score computed. |
| Baseline fingerprint includes a safety car lap | Lap flag check during baseline computation | Exclude flagged lap. If fewer than 3 clean laps available for baseline: extend window to 5 laps. If still insufficient: return BASELINE_INSUFFICIENT. |
| Driver historical variation data unavailable (new circuit) | Lookup in /artefacts/stint_variation.json | Use global driver variation as fallback. Tag alert with FALLBACK_THRESHOLD. Report in UI. |
| All 3 baseline laps have INTERPOLATED flag | Check INTERPOLATED tag on baseline laps | Accept baseline but tag stint monitor as DEGRADED_QUALITY. Alert confidence capped at LOW regardless of deviation score. |
| Cosine distance returns NaN | NaN check on every distance computation | Skip lap in rolling mean. Log DISTANCE_NAN. If 3 consecutive NaN results: return MONITOR_ERROR and halt monitoring for this stint. |

# 6. Explainability Architecture

Every APEX prediction is accompanied by a full SHAP decomposition. Explainability is not an add-on — it is a first-class output. The animated UI is built around SHAP values. An unexplained prediction is not a valid APEX output.

## 6.1 SHAP Implementation

| | |
|---|---|
| **Library** | SHAP 0.45+. TreeExplainer for XGBoost models (exact, not approximate). |
| **Scope** | Applied to Model 1 (classifier) for every prediction. Applied to personalisation engine for every setup recommendation. |
| **Global explainability** | Beeswarm plot: each dot is a lap, x-axis is SHAP value, colour is feature value. Computed once after training and stored as artefact. |
| **Local explainability** | Force plot: for each lap prediction, which features pushed toward or away from the predicted driver. Stored in PostgreSQL as JSONB per prediction. |
| **Radar derivation** | SHAP values aggregated across the 6 feature domains (throttle, braking, speed, gear, engine, cornering) per driver. These 6 domain scores form the 6-axis fingerprint radar displayed in the UI. |
| **Multi-season stability** | SHAP distributions compared across 2022, 2023, 2024 using Jensen-Shannon divergence. Quantifies how much each driver's fingerprint drifted year-on-year. |
| **Natural language generation** | SHAP domain scores converted to natural language descriptions using a template engine. Top 2 domains by magnitude generate the style description. Example: 'This driver is characterised by aggressive late braking with high throttle-brake overlap in slow corners.' |
| **Computation time** | TreeExplainer SHAP for one lap: < 50ms. For batch of 100 laps: < 2 seconds. Computed synchronously for single-lap predictions. Batch SHAP runs as Celery task. |

## 6.2 SHAP Failure Handling

| Failure Mode | Response |
|---|---|
| SHAP computation raises exception (e.g. model version mismatch) | Return prediction without SHAP. Tag response with SHAP_UNAVAILABLE. Log full traceback. Do not suppress prediction. |
| SHAP values sum to value far from expected (model consistency check fails) | Log SHAP_CONSISTENCY_WARN. Return SHAP values with warning flag. Include in monitoring dashboard. |
| Natural language template finds no dominant domain (all scores within 10% of each other) | Return generic description: 'This driver shows balanced characteristics across all domains with no single dominant style trait.' Do not fabricate a dominant trait. |

# 7. Car Personalisation Engine

The personalisation engine translates a driver fingerprint into car setup recommendations. It operates in three layers: a learned mapping function, a rule-based constraint layer, and a confidence scoring layer. No recommendation reaches the user without passing all three layers.

## 7.1 Layer 1 — Learned Mapping (XGBoost Regression per Parameter)

One XGBoost regressor is trained per setup parameter. Input: driver fingerprint 5D vector + circuit type (3D one-hot). Output: predicted optimal setup value for that parameter. Training data: best-performing stints per driver per circuit, with setup values approximated from telemetry signatures.

> **⚠ Setup values are approximated, not measured directly. Brake bias is inferred from longitudinal G distribution. Front wing level is inferred from speed delta in high-speed corners vs slow corners. These approximations are explicitly stated as limitations in the dissertation. The system provides a starting direction, not a precise engineering specification.**

## 7.2 Layer 2 — Constraint Engine

Every recommendation is passed through a constraint layer before returning to the user. Constraints are hard limits — they override the learned mapping output.

| Constraint | Rule | Override Behaviour |
|---|---|---|
| FIA regulation bounds | Each parameter has a legal min/max per season. Values outside this range are clipped to the boundary. | Clip to boundary. Log CONSTRAINT_CLIPPED with original and clipped value. |
| Physical consistency: front wing vs brake bias | Extreme front wing (>10 degrees) combined with very low brake bias (<54%) is mechanically inconsistent for most circuits. | If both conditions met: flag CONSISTENCY_WARNING in response. Do not override — flag for engineer review. |
| Monaco hard constraint | Monaco always recommends maximum downforce regardless of fingerprint. Circuit-specific override. | Override front and rear wing to maximum legal value. Log CIRCUIT_OVERRIDE_MONACO. |
| Confidence floor | If the learned mapping regression confidence is < 0.5 for any parameter: return range, not point estimate. Range = ±15% of predicted value. | Return range. Flag parameter with WIDE_CONFIDENCE in response. |

## 7.3 The 10-Parameter Setup Vector

| Parameter | Fingerprint Features Driving It | Output Format | FIA Bounds (approx) |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Front wing angle | F19 (lateral_g_mean), F14 (speed_p90) | Degrees + confidence | 0–12 degrees |
| Rear wing angle | F04 (throttle_full_fraction), F14 | Degrees + confidence | 0–15 degrees |
| Brake bias | F06, F08, F10 (brake features) | % front + confidence | 52–62% |
| Differential ramp entry | F25 (corner_entry_balance), F09 | Index (1–10) + confidence | Team-defined |
| Differential ramp exit | F03 (throttle_open_rate), F22 | Index (1–10) + confidence | Team-defined |
| Throttle map curve | F01, F02, F03, F05 | Curve index (1–9) + confidence | 1–9 |
| Suspension stiffness front | F21 (steering_correction_rate), F19 | Soft / Medium / Stiff | Team-defined |
| Suspension stiffness rear | F12 (speed_variance), F20 | Soft / Medium / Stiff | Team-defined |
| Tyre warm-up laps estimate | F27 (tyre_age_throttle_slope) | Integer laps (1–5) | N/A |
| Brake duct aperture | F06, F07, F08 | % open + confidence | 30–100% |

## 7.4 Unknown Driver Onboarding Output

When an unknown driver's telemetry is uploaded, the personalisation engine produces a complete onboarding package as if they were joining a new team:

- Style match report: top 3 nearest known F1 drivers in 5D style space with cosine similarity scores and confidence bounds
- Setup recommendation sheet: all 10 parameters with confidence ratings (HIGH / MEDIUM / LOW / WIDE)
- Fingerprint radar: 6-axis chart with natural language style description
- Risk flags: feature dimensions where the unknown driver's fingerprint falls outside the distribution of all known F1 drivers — potential adaptation challenges flagged explicitly
- Data quality summary: how many laps were used, how many rejected, which features were imputed

# 8. System Architecture

## 8.1 Overview

APEX is a single deployable backend application with clearly separated internal modules, a Next.js frontend, a PostgreSQL database, a Redis cache, and a Celery task queue for async ML jobs. The architecture is designed for local development reliability first and cloud deployment second. Every component has a defined health check.

## 8.2 Application Modules

| Module | Location | Responsibility | Failsafe Behaviour |
|---|---|---|---|
| Ingestion | app/pipeline/ingest.py | FastF1 and OpenF1 data pull, caching | Retry 3x with exponential backoff. After 3 failures: return DATA_SOURCE_UNAVAILABLE. Use cached data if available and < 7 days old. |
| Validator | app/pipeline/validate.py | Lap filter chain, error code assignment, cleaning report generation | Never raises unhandled exception. All errors caught, coded, and returned in ValidationResult object. |
| Feature Extractor | app/pipeline/features.py | 28-feature computation per lap, null handling, range validation | Each feature computed in isolation. One feature failure does not block others. Null tolerance policy enforced. |
| Normaliser | app/pipeline/normalise.py | Circuit-aware Z-score normalisation using saved training parameters | If normalisation params missing for a circuit: use global params. Log NORMALISATION_FALLBACK. |
| Classifier | app/ml/classifier.py | XGBoost inference, confidence gating, SHAP computation | Confidence gate enforced on every call. SHAP failure does not block prediction. |
| Style Space | app/ml/stylespace.py | UMAP transform for known and unknown drivers | PCA fallback if UMAP fails. Out-of-distribution check on every projection. |
| Degradation Monitor | app/monitoring/detector.py | Within-stint deviation scoring, alert generation | Minimum lap count enforced. Baseline quality checks. NaN-safe distance computation. |
| Personalisation | app/personalisation/engine.py | Setup recommendation, constraint layer, | Constraint layer always runs. Wide confidence returned on low-confidence inputs. |

| | | | |
|---|---|---|---|
| | | confidence scoring | |
| Cache | app/cache/redis.py | Driver profile cache, normalisation param cache, model output cache | If Redis unavailable: all cache calls return CACHE_MISS gracefully. System operates without cache. Log CACHE_UNAVAILABLE. |
| Task Queue | app/tasks/celery.py | Async batch SHAP computation, large session ingestion | If Celery worker unavailable: synchronous fallback for single-lap requests only. Batch requests queued until worker recovers. |

## 8.3 API Circuit Breakers

External API calls (FastF1, OpenF1, Ergast) are wrapped in a circuit breaker pattern. After 3 consecutive failures to the same endpoint within a 60-second window, the circuit opens and subsequent calls return immediately with a cached response or a defined fallback — no hanging requests.

| External Dependency | Timeout | Retry Policy | Circuit Breaker | Fallback |
|---|---|---|---|---|
| FastF1 session fetch | 30s | 3 retries, 5s / 15s / 30s backoff | Opens after 3 failures in 60s | Return cached Parquet if available. Else DATA_SOURCE_UNAVAILABLE. |
| OpenF1 REST API | 10s | 3 retries, 2s / 5s / 10s backoff | Opens after 3 failures in 60s | Use Ergast API. If both fail: COMPOUND_UNKNOWN flag. |
| Ergast API | 10s | 2 retries, 3s / 8s backoff | Opens after 2 failures in 60s | Derive tyre age from pit flags only. |
| Supabase PostgreSQL | 5s | 3 retries, 1s / 2s / 4s backoff | Opens after 3 failures in 30s | Return DB_UNAVAILABLE. API returns 503. No silent data loss. |
| Redis Cache | 2s | 1 retry, 1s backoff | Opens after 2 failures in 30s | Cache miss on all reads. Writes silently dropped. Log CACHE_DEGRADED. |

## 8.4 Database Schema

| Table | Key Columns | Purpose | Integrity Constraints |
|---|---|---|---|
| drivers | id UUID PK, code VARCHAR(3), full_name, team, season | Driver master data | UNIQUE(code, season). 8 known + unlimited unknown drivers. |
| circuits | id UUID PK, name, type ENUM(street, | Circuit master data | UNIQUE(name). 10 circuits. |

| | | | |
|---|---|---|---|
| | high_speed, technical, mixed) | | |
| sessions | id UUID PK, circuit_id FK, season, round, session_type | Race session records | UNIQUE(circuit_id, season, round, session_type). |
| laps | id UUID PK, session_id FK, driver_id FK, lap_number, is_clean BOOL, error_code, tyre_compound, tyre_age, has_interpolation BOOL | Per-lap metadata with quality flags | NOT NULL on is_clean and error_code. error_code='OK' for clean laps. |
| lap_features | id UUID PK, lap_id FK UNIQUE, feature_vector JSONB, null_count INT, imputed_features TEXT[], corner_type ENUM | 28-feature vectors | null_count must equal count of null values in feature_vector. Enforced by application layer. |
| fingerprints | id UUID PK, driver_id FK, season, circuit_id, vector JSONB(5D), shap_global JSONB, radar_values JSONB(6D), stability_score FLOAT | Aggregate fingerprint per driver/season/circuit | UNIQUE(driver_id, season, circuit_id). |
| predictions | id UUID PK, lap_id FK, predicted_driver_id FK, confidence FLOAT, confidence_state ENUM, shap_values JSONB, shap_available BOOL | Classifier output | confidence_state must be HIGH/MEDIUM/LOW/AMBIGUOUS. No NULL confidence. |
| setup_recommendations | id UUID PK, driver_id FK, circuit_id FK, setup_vector JSONB, confidence_vector JSONB, constraint_flags | Personalisation output | constraint_flags records any constraint layer overrides. |

| | | | |
|---|---|---|---|
| | JSONB, model_version VARCHAR | | |
| degradation_alerts | id UUID PK, session_id FK, driver_id FK, lap_number, deviation_score FLOAT, alert_confidence ENUM, feature_attribution JSONB, lead_time_actual INT | Degradation monitor output | lead_time_actual populated post-race for validation. |
| data_quality_log | id UUID PK, created_at, entity_type, entity_id, error_code, error_detail, fallback_used BOOL | All pipeline errors and fallbacks | Every DATA_, LAP_, UPLOAD_, COMBO_, FEATURE_ error logged here. Never deleted. |

## 8.5 API Endpoints

| Method | Endpoint | Response Contract | Error States |
|---|---|---|---|
| GET | /health | { status, models_loaded, db_connected, cache_connected, version } | 503 if any critical component unavailable |
| GET | /drivers | Array of driver summaries with fingerprint radar values | 200 always — returns empty array if no data |
| GET | /drivers/{id}/fingerprint | Full fingerprint: 5D vector, SHAP radar, natural language description, multi-season stability | 404 if driver not found. 206 if partial data (some seasons missing). |
| GET | /drivers/{id}/setup/{circuit_id} | Setup recommendation with 10 parameters, confidence ratings, constraint flags | 404 if no data for combination. 200 with WIDE_CONFIDENCE if limited data. |
| GET | /drivers/{id}/stability | Multi-season JS divergence scores, year-on-year fingerprint comparison | 206 if fewer than 3 seasons available. |
| POST | /predict | { predicted_driver, confidence, confidence_state, top_3, shap_values, shap_available } | 400 on invalid input. 200 with LOW_CONFIDENCE state — never 500 on model output. |
| POST | /ingest/session | { session_id, laps_accepted, laps_rejected, rejection_summary, quality_report } | 202 Accepted — processing is async. |

| | | | |
|---|---|---|---|
| | | | Poll /ingest/status/{session_id}. |
| GET | /ingest/status/{session_id} | { status: PENDING/PROCESSING/COMPLETE/FAILED, progress, error_detail } | 404 if session_id unknown. |
| POST | /ingest/upload | { upload_id, laps_accepted, laps_rejected, cleaning_report, warnings } | 400 on UPLOAD_001 through UPLOAD_004. 200 with warnings for non-fatal issues. |
| POST | /fingerprint/unknown | { fingerprint_id, fingerprint_vector, radar_values, style_description, nearest_neighbours, risk_flags, data_quality } | 200 with SPARSE_PROJECTION warning if <30 laps. 400 if <5 laps. |
| GET | /compare/{id1}/{id2} | { driver_1_radar, driver_2_radar, delta_shap, most_distinctive_features, compatibility_delta } | 404 if either driver not found. |
| GET | /stylespace | { drivers: [{id, code, coords_2d, coords_5d, season}], unknown_drivers: [...] } | 200 always — empty arrays if no data. |
| GET | /replay/{session_id}/{driver_id} | { laps: [{lap_number, feature_vector, deviation_score, alert_state, fingerprint_snapshot}] } | 404 if session or driver not found. 206 if partial data. |
| GET | /metrics | Prometheus-format metrics: request count, latency, cache hit rate, model confidence distribution | Internal only. 403 if called externally. |

# 9. Technology Stack

| Layer | Technology | Version | Purpose & Failsafe Note |
|---|---|---|---|
| Language — backend | Python | 3.11 | All pipeline, ML, and API code. 3.11 for performance and typing. |
| Language — frontend | TypeScript | 5.x | All frontend. Strict mode enabled. No implicit any. |
| API Framework | FastAPI | 0.111+ | Async REST. Pydantic v2 for all request/response validation. Invalid inputs rejected at schema layer before reaching business logic. |
| Database | PostgreSQL via Supabase | 15 | Primary store. JSONB for vectors and SHAP arrays. Row-level security enabled. |
| ORM | SQLAlchemy 2.0 + asyncpg | 2.0 | Async DB access. Alembic for migrations. All schema changes versioned. |
| Cache | Redis via Upstash | 7.x | Driver profile cache. 1-hour TTL. System operates fully without cache — cache is performance optimisation, not dependency. |
| Task Queue | Celery + Redis broker | 5.x | Async ML batch jobs. Synchronous fallback for single-lap requests if worker unavailable. |
| Data ingestion | FastF1 | 3.x | Official F1 telemetry. Local Parquet cache after first download. Cache used if source unavailable. |
| Context data | OpenF1 REST API | v1 | Tyre compound, pit flags. Ergast fallback if unavailable. |
| ML — Classification | XGBoost | 2.x | 8-class driver classifier. CalibratedClassifierCV for probability calibration. |
| ML — Hyperparameter | Optuna | 3.x | Bayesian search. SQLite storage backend for trial persistence across restarts. |
| ML — Style Space | UMAP-learn | 0.5.x | Driver style space. PCA maintained as fallback artefact. |
| ML — Explainability | SHAP | 0.45+ | TreeExplainer. Exact (not approximate) for XGBoost. Async batch computation. |
| Data processing | pandas 2.x + numpy 1.x | 2.x / 1.x | Feature extraction. All operations vectorised — no Python loops over rows. |
| Experiment tracking | MLflow | 2.x | All training runs logged. Champion model registered. Reproducibility via logged seeds and configs. |
| Frontend framework | Next.js 14 | 14.x | App Router. React 18. SSR for initial fingerprint data, CSR for animations. |
| Animation | D3.js | 7.x | All animated charts: radar, force plot, beeswarm, style scatter, deviation timeline. |

| | | | |
|---|---|---|---|
| 3D Visualisation | Three.js | r128 | 3D style space map. Documented D3.js 2D fallback if Three.js integration fails. |
| State management | Zustand | 4.x | Global UI state. Driver selection, circuit filter, replay position. |
| Styling | Tailwind CSS | 3.x | Dark theme. Design tokens in tailwind.config.js. No inline styles. |
| Containerisation | Docker + Compose | 24.x | All components containerised. docker-compose.yml for local. Single command startup. |
| CI/CD | GitHub Actions | — | On every push: run tests, check model accuracy gate (macro-F1 must remain > 0.75 or build fails), lint, type check. |
| Backend hosting | Railway | — | Push-to-deploy. Auto TLS. Env vars in Railway dashboard. |
| Frontend hosting | Vercel | — | Next.js native deployment. Environment-specific config via Vercel env vars. |
| Monitoring | Prometheus + Grafana | — | API latency, cache hit rate, prediction confidence distribution, circuit breaker state. |
| Dependency management | Poetry (Python) | 1.8+ | Locked dependencies via poetry.lock. Checked into version control. |

# 10. Animated UI Specification

APEX has no static pages. Every screen is live. The visual language is dark, high-contrast, and precision-engineered — referencing F1 TV data overlays and MoTeC displays while adding a layer of interactivity those tools lack. Every number that changes counts up or down with a tween. Every chart animates in on load. Every state change is a transition, not a snap.

## 10.1 Global Design System

| | |
|---|---|
| **Background** | #0D0D0D — near-black, consistent across all screens |
| **Primary accent** | #E8002D — F1 red. Used for headings, active states, alert indicators. |
| **Secondary accent** | #C0392B — deep red. Used for SHAP force values and warnings. |
| **Text primary** | #F0F0F0 — near-white |
| **Text secondary** | #888888 — mid-grey for labels and metadata |
| **Card background** | #1A1A1A — slightly lighter than background |
| **Border** | #2B2B2B — subtle separation |
| **Font** | Inter (UI text), JetBrains Mono (telemetry values, lap numbers, feature values) |
| **Animation easing** | D3 easeCubicInOut for all chart transitions. Duration 600ms default, 1200ms for style space transitions. |
| **Loading state** | Animated skeleton screens with a telemetry-wave SVG pattern. No spinners. |
| **Error state** | Red left border on affected card. Error code shown in monospace. Retry button where applicable. |

## 10.2 Screen 1 — Command Centre

The landing screen. An 8-driver grid where every card is alive. Each card shows the driver code, team colour bar, a small animated 6-axis radar chart (the fingerprint), current circuit compatibility score, and fingerprint stability indicator across seasons.

- On page load: all 8 radar charts animate simultaneously — each axis draws outward in sequence with a 80ms stagger between axes, creating a bloom effect. Duration: 800ms total.
- Circuit selector in top navigation: changing circuit triggers all 8 radars to transition to the new circuit's fingerprint simultaneously. Duration: 600ms.
- Hover on driver card: card expands inline (no navigation) showing top 3 most distinctive SHAP features as horizontal bar charts animating outward from zero.
- Confidence state indicator: if a driver's fingerprint has LOW_CONFIDENCE flag, their card shows a dashed border and a warning badge. The radar still displays but with reduced opacity.
- Click: card-expand transition — card scales and fades to fill the screen, revealing the Driver Profile.

- Sort controls: sort by compatibility score, fingerprint stability, or degradation sensitivity. Cards re-order with a layout animation.

## 10.3 Screen 2 — Driver Profile

The full profile for one driver. Split layout: left panel shows the animated radar and fingerprint metadata. Right panel shows the SHAP beeswarm and season timeline.

- Large 6-axis radar: each axis labelled with domain name and current score. Tooltip on hover: domain name, score, natural language meaning, comparison to field mean. Radar animates on load — axis by axis, 100ms stagger.
- Season switcher: three tabs (2022, 2023, 2024). Switching season animates the radar to the new fingerprint. A ghost of the previous season's radar remains visible in reduced opacity for 2 seconds before fading.
- SHAP beeswarm: dots animate in from the centre line, each dot a lap, colour coded by circuit. Hover shows lap number, session, predicted confidence.
- Multi-season overlay mode: toggle overlays all three seasons on one radar. Each season a different line style (solid, dashed, dotted). JS divergence score displayed between each pair.
- Corner type tabs: Slow / Medium / Fast. Clicking re-animates the radar for that corner class sub-fingerprint.
- Natural language description: auto-generated from SHAP domain scores. Updates when season or corner type changes. Text fades out and fades in on change.
- Setup recommendation panel below radar: 10 parameters each with an animated gauge needle settling to the recommended value on load. Wide confidence parameters show a range bar instead of a point needle.

## 10.4 Screen 3 — Ingestion & Prediction

Three tabs: FastF1 Session, Manual Upload, Unknown Driver. This screen shows the system's own thinking — the pipeline stages animate as they complete.

### All tabs — Pipeline Progress Animation

- On submit: a horizontal progress bar appears divided into named stages: Fetching → Validating → Extracting Features → Running Model → Computing SHAP.
- Each stage lights up green as it completes. If a stage fails: it turns red with the error code displayed inline. Subsequent stages do not proceed.
- Stage timing shown in monospace next to each stage label (e.g. 'Extracting Features  2.3s').

### FastF1 Session Tab

- Dropdowns: Season → Round → Driver → Session Type. Each dropdown populates based on the previous selection (dependent dropdowns).
- 'Laps available' count updates dynamically as round and driver are selected, showing how many clean laps are expected.
- Result: predicted driver label animates in. Confidence bar fills left to right with colour coded by state (green > 0.80, amber 0.65–0.80, red < 0.65 — shows LOW_CONFIDENCE state).

- SHAP force plot: feature bars extend outward from baseline, red (toward prediction) and blue (away). Each bar animates from zero. Hover shows feature name, value, and SHAP contribution.

### Manual Upload Tab

- Drag-and-drop CSV zone. Animated dashed border pulses while file is being read.
- On upload: cleaning report animates in as a table. Accepted laps count up. Rejected laps count up. Rejection reasons shown per lap with error codes.
- Warnings (e.g. UPLOAD_002 unit warning) appear as amber inline alerts with dismiss button.

### Unknown Driver Tab

- Upload with no label. System generates fingerprint.
- Radar animates axis by axis as each feature domain is computed (streamed updates via server-sent events).
- Style space opens in split view (Screen 5 embedded) showing where the unknown driver lands. Their marker animates in — drifting from the origin to its position over 1.2 seconds.
- Nearest-neighbour matches animate in below: top 3 known drivers with similarity scores shown as filled progress bars.
- Risk flags shown as red warning cards with feature names and 'outside known F1 driver distribution' label.

## 10.5 Screen 4 — Comparison View

Two drivers selected side by side. The most useful screen for a race engineer.

- Two radars rendered side by side. Both animate on load simultaneously.
- Delta radar: a third radar showing the signed difference between the two drivers. Positive deltas in red (Driver 1 higher), negative in blue (Driver 2 higher).
- Most-distinctive features: top 5 features by absolute SHAP delta, shown as horizontal bar chart. Bars animate from zero. Hover shows plain English meaning of the difference.
- Swap button: drivers swap positions. Radars cross-fade to swapped positions. Delta radar flips sign.
- Compatibility projection: circuit selector allows engineer to ask 'given Car X's current setup for this circuit, which driver gets better compatibility?' Animated gauge shows the delta.
- Setup delta: which setup parameters differ most between the optimal setups for each driver? Shown as an animated difference table.

## 10.6 Screen 5 — Style Space Map

Three.js 3D scatter plot of all drivers in style space (3 of 5 UMAP dimensions rendered). The most visually striking screen in the system.

- 8 driver spheres, each glowing in team colour. Auto-rotating camera at 0.2 deg/second. User can click and drag to control camera.
- Hover: sphere pulses. Tooltip appears showing driver code, team, season, nearest neighbours, and mini radar.

- Click: camera animates to face the selected driver's sphere. Sidebar opens with their fingerprint summary.
- Unknown driver projection: when an unknown driver is profiled, a white sphere animates in — appearing at the origin and drifting to its position over 1.5 seconds. Label: 'Unknown Driver'.
- Teammate proximity lines: toggle shows connecting lines between teammates. Line thickness encodes intra-team cosine distance — thicker = more similar.
- Season filter: selecting a single season moves all spheres to their season-specific positions. Transition duration 1.2 seconds with cubic easing.
- D3.js fallback: if Three.js fails to initialise (WebGL unavailable), system silently falls back to a D3.js 2D scatter plot with the same interactions. User sees no error — just a 2D view.

## 10.7 Screen 6 — Historical Race Replay Monitor

Historical race replay. The user selects a session and a driver. The system replays the race lap by lap, showing the fingerprint evolving in real time as if monitoring a live race. No actual live data dependency — fully controlled and reliable for demos and vivas.

- Playback controls: Play, Pause, Speed (1x, 5x, 10x), Scrub bar showing lap number.
- Deviation score gauge: animated arc gauge that updates every lap as playback advances. Colour transitions from green (low deviation) to amber to red (alert threshold exceeded).
- Fingerprint radar: updates every lap as playback advances. Previous lap's radar shown as ghost for 2 seconds before fading.
- Tyre deg timeline: horizontal chart building right to left as laps are replayed. Deviation score plotted as a line. Alert threshold shown as a horizontal dashed red line. When the line crosses the threshold, the chart pulses red.
- Alert banner: when degradation alert fires, a full-width red banner animates down from the top of the screen showing: alert confidence, feature drivers, laps since threshold crossed, recommended pit window.
- 8-driver mode: replay all 8 drivers simultaneously in a grid. Each driver has their own mini deviation gauge and tyre timeline. Alert banners per driver.

# 11. Evaluation Framework

Every evaluation in APEX reports failure cases alongside success metrics. A system that only shows what it gets right is not an honest system. The dissertation will dedicate equal space to where APEX fails, why it fails, and what that implies for the research questions.

## 11.1 Model 1 — Classifier Evaluation

| Metric | Target | Reported On | Failure Analysis |
|---|---|---|---|
| Macro-averaged F1 | > 0.78 | Held-out circuits (Belgium, Italy) | If below target: report which drivers are confused most. Likely teammates (similar car, similar circuits). Analyse whether confusion is meaningful or artificial. |
| Per-driver F1 | > 0.70 for all 8 drivers | Held-out circuits | Any driver below 0.70 is a research finding: report why. Hypothesise whether it is a style problem or a data problem. |
| Confusion matrix | Visual heatmap | Held-out circuits | Systematic confusions (e.g. VER-PER as Red Bull teammates) are expected and reported as a finding, not a failure. |
| Calibration curve | Predicted probabilities close to empirical frequencies | Test set | Poorly calibrated model means confidence scores are unreliable. If miscalibrated: report and adjust confidence gate threshold accordingly. |
| Baseline comparison | XGBoost must exceed all 3 baselines | Training + held-out | If XGBoost fails to exceed Random Forest: analyse which features Random Forest uses more effectively. |
| Normalisation ablation | Without normalisation: F1 drops significantly (expected) | Training circuits only | Quantifies how much of model performance is driver style vs circuit identity. Key dissertation result. |

## 11.2 Fingerprint Stability Evaluation

| Metric | Target | Method | Failure Analysis |
|---|---|---|---|
| Year-on-year stability (JS divergence) | < 0.15 for drivers with stable known style (e.g. Verstappen) | JS divergence on SHAP distributions: 2022 vs 2023, 2023 vs 2024 | High divergence for a driver is a research finding: did their style change, or did the car change? Discuss with reference to known team performance shifts. |
| Teammate proximity | Intra-team distance < inter-team distance | Mean Euclidean distance in 5D space | If test fails for a team: analyse whether car characteristics dominate driver style for that team's |

| | | | |
|---|---|---|---|
| | | | configuration. Discuss car-driver confound. |
| Circuit generalisation | F1 > 0.70 on Belgium and Italy | Held-out circuit evaluation | Report which features degrade most on unseen circuits. Hypothesise: are these circuit-specific or driver-specific signals? |
| Unknown driver projection | Correct nearest-neighbour found for 6/8 drivers acting as unknowns | Re-insert each known driver as unknown, check nearest-neighbour in style space | Failures reported: which drivers are projected to incorrect nearest neighbours and why. |
| UMAP stability | Procrustes-aligned variance < 0.05 across 10 random seeds | 10 UMAP runs, Procrustes alignment | High variance means style space is unstable. Report and discuss implications for unknown driver projection reliability. |

## 11.3 Degradation Detector Evaluation

| Metric | Target | Method | Failure Analysis |
|---|---|---|---|
| Mean alert lead time | 2–5 laps before pit | All 2023–2024 race stints for 8 drivers. Pit lap as ground truth. | Negative lead time (alert after pit) means system missed the signal. Report which circuits and compounds produce most misses. |
| True positive rate | > 70% of stints alerted within 5 laps before pit | Per-stint binary classification (alerted on time / not) | Report false negative stints: were they short stints, unusual compounds, or safety car affected? |
| False positive rate | < 20% of stints alert more than 5 laps before pit | Per-stint analysis | Early alerts reduce engineer trust. Report which driver-circuit combinations produce most early alerts. |
| Compound stratification | Reported separately for Soft, Medium, Hard, Intermediate | Stratified analysis by compound | Expected: Soft compound shows clearest signal (fastest deg). Hard compound may show no reliable signal. Report this honestly. |
| Threshold sensitivity analysis | Report TP rate at 1SD, 1.5SD, 2SD, 2.5SD thresholds | Vary threshold, measure TP/FP tradeoff | Report optimal threshold per circuit type. Discuss implications for a real race engineer calibrating the system. |

## 11.4 Failsafe System Evaluation (RQ5)

This evaluation is unique to APEX and directly addresses RQ5. No prior published system has formally evaluated the failure modes of an F1 telemetry ML system. This section is a dissertation contribution in itself.

| Test | Method | Expected Result |
|---|---|---|
| Confidence gate effectiveness | Inject deliberately ambiguous telemetry (averaged across 2 drivers). Measure rate of LOW_CONFIDENCE states returned. | System should return LOW_CONFIDENCE for ≥90% of ambiguous inputs rather than a wrong confident prediction. |
| Circuit breaker activation | Simulate FastF1 API unavailability by mocking 3 consecutive failures. Verify circuit opens and cached data is returned. | No uncaught exception. System returns cached data with STALE_DATA flag within 500ms. |
| NULL feature tolerance | Inject lap vectors with 1, 3, 6, 7 NULL features. Verify correct accept/reject behaviour at boundary. | Vectors with ≤6 NULLs accepted with imputation. Vectors with 7+ NULLs rejected with LAP_NULL_EXCESS. No silent imputation. |
| Cache unavailability resilience | Disable Redis. Run 100 consecutive predictions. Measure accuracy and response time degradation. | Zero failed predictions. Response time increases by < 200ms average. Log shows CACHE_UNAVAILABLE throughout. |
| Database unavailability | Simulate DB timeout after 3 retries. Verify 503 response and no data loss. | API returns 503 with DB_UNAVAILABLE reason. No partial writes. No unhandled exceptions. |
| UMAP out-of-distribution | Inject synthetic telemetry with feature values 5 standard deviations outside training distribution. Verify OUT_OF_DISTRIBUTION warning. | Warning returned. Projection computed but flagged. UI shows dashed marker. No crash. |
| Concurrent request handling | Fire 50 concurrent prediction requests. Verify no race conditions, consistent results, acceptable latency. | All 50 requests complete. Results consistent (same input = same output). p95 latency < 500ms. |

# 12. Future Integration Specification

APEX is designed as the behavioural intelligence layer of a larger race strategy platform. The following defines the integration contract for the post-submission integration with a pit stop and ERS optimisation system.

## 12.1 APEX Outputs Exposed to Partner System

| | |
|---|---|
| **Driver fingerprint vector** | GET /drivers/{id}/fingerprint → 5D vector. Refreshed per session. Used by ERS system as driver profile input. |
| **Compatibility score** | GET /drivers/{id}/setup/{circuit_id} → compatibility_score (0–1). Used by strategy system to weight setup-dependent performance predictions. |
| **Degradation signal** | GET /replay/{session_id}/{driver_id} → deviation_score per lap. Partner system consumes this as input to pit window optimisation. |
| **Tyre window estimate** | Included in degradation alert payload: optimal_window_remaining_laps (int). Direct input to pit stop timing model. |
| **Stint performance baseline** | Per-driver expected lap time consistency score for a given setup and circuit. Used to calibrate strategy model expectations. |

## 12.2 Integration Architecture

| | |
|---|---|
| **APEX internal port** | 8001 |
| **Partner system internal port** | 8002 |
| **Shared database** | Separate schemas in same PostgreSQL instance: apex_schema and strategy_schema |
| **API gateway** | Nginx reverse proxy at port 80. Routes /apex/* to 8001 and /strategy/* to 8002 |
| **Authentication** | JWT issued by shared auth service. Both systems validate the same token. |
| **Deployment** | Single docker-compose.yml with APEX and partner system as separate services on shared Docker network |

# 13. Project Plan — 22 Weeks

Redistributed from original plan. CVAE removed (saves 3 weeks). Live WebSocket replaced with replay engine (saves 1.5 weeks). Freed weeks redistributed to hardening, failsafe testing, and evaluation. Core system complete and tested by Week 14. UI complete by Week 19. Full evaluation and dissertation by Week 22.

| Week | Phase | Task | Deliverable | Risk |
|---|---|---|---|---|
| 1 | Foundation | Environment setup. FastF1 cache validation for all 10 circuits × 8 drivers × 3 seasons. Verify minimum 30 clean laps per driver-circuit combination. | Data coverage report. List of any combinations below minimum threshold. | FastF1 gaps. Drop combination if below threshold. |
| 2 | Foundation | Lap filter chain implementation. All 9 filter rules coded. Unit tests for each filter with synthetic passing and failing laps. | Validated filter pipeline. 100% test coverage on filter functions. | Low. |
| 3 | Foundation | Feature extraction pipeline. All 28 features coded. Unit tests for every feature function including range validation and NULL trigger conditions. | feature_extraction.py with full test suite. Feature computation contract documented. | Medium — some features (F19–F21) require X/Y data which may be patchy. |
| 4 | Foundation | Circuit normalisation. Normalisation parameters computed on training circuits only. Saved to artefacts. Ablation test setup prepared. | normalisation.json artefact. Normalisation leakage test passing. | Low. |
| 5 | Core ML | XGBoost baseline training. MLflow setup. Optuna study initialised. Baseline comparisons (majority class, logistic regression, random forest) run and logged. | MLflow experiment with baseline runs. Baseline F1 scores documented. | Low. |
| 6 | Core ML | Optuna hyperparameter search. Minimum 200 trials. Champion model registered. Calibration curve computed. Confidence gate threshold set. | Champion XGBoost model in MLflow registry. Calibration plot. Confidence gate validated. | Medium — F1 may not hit 0.78 first pass. Feature engineering revision may be needed. |

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| 7 | Core ML | SHAP computation on full test set. Global beeswarm. Per-driver force plots. Radar domain aggregation. Natural language template engine. | SHAP artefacts stored. Radar values for all 8 drivers. NL descriptions generated. | Low. |
| 8 | Core ML | UMAP style space. 5D and 2D projections. Stability test (10 seeds, Procrustes). Teammate proximity test. PCA fallback artefact created. | UMAP model artefact. Stability report. Proximity test results. | Low. |
| 9 | Core ML | Corner type decomposition. Curvature computation from X/Y. Per-corner feature sub-vectors. Ablation: does corner-type information improve classifier? | Corner sub-fingerprints for all drivers. Ablation result. | Medium — X/Y data quality varies. Failure handling must be in place. |
| 10 | Hardening | Full pipeline integration test. Run complete pipeline end-to-end on 3 circuits. Verify error codes propagate correctly. Verify data_quality_log populated. | Integration test suite passing. Quality log populated correctly. All error codes tested. | Medium. |
| 11 | Intelligence | Degradation detector. Baseline fingerprint computation. Deviation score. Rolling mean. Alert threshold calibration from training data. | Degradation alerts on 2022 validation stints. Threshold values per driver per circuit type in artefacts. | Medium. |
| 12 | Intelligence | Degradation validation. Lead time computation against known pit laps in 2023–2024 data. Compound stratification. Threshold sensitivity analysis. | Validation results table. Lead time metric. Sensitivity analysis plot. | Medium. |
| 13 | Intelligence | Personalisation engine. Mapping function training (one regressor per setup parameter). Constraint layer. Confidence scoring. | Setup recommendations for all 8 drivers × 10 circuits. Constraint flags verified. | Medium. |
| 14 | Hardening | Failsafe system evaluation (RQ5). All 7 failsafe tests run. Circuit breaker tests. NULL | RQ5 evaluation results table. All 7 tests passing or | Low — these are controlled tests. |

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| | | tolerance boundary tests. Cache unavailability test. Concurrent request test. | documented as findings. | |
| 15 | Platform | PostgreSQL schema finalised. Alembic migrations. FastAPI skeleton. All endpoints stubbed with Pydantic schemas. Health endpoint returning real component status. | Running API. All endpoints returning correct schemas. Health check passing. | Low. |
| 16 | Platform | FastAPI full implementation. Redis cache layer. Celery task queue. Replay engine (lap-by-lap historical data serving). All error states returning correct HTTP codes. | All 14 endpoints functional and tested. Replay engine returning lap-by-lap data. | Medium. |
| 17 | Platform | Next.js setup. Design system implemented in Tailwind. Screen 1 (Command Centre) and Screen 2 (Driver Profile) fully animated. | Two screens animated and connected to live API. | Medium. |
| 18 | Platform | Screen 3 (Ingestion), Screen 4 (Comparison), Screen 5 (Style Space with Three.js). D3.js fallback for Screen 5 implemented and tested. | Screens 3–5 animated. Three.js working. D3 fallback tested. | High — Three.js most complex UI component. |
| 19 | Platform | Screen 6 (Historical Replay). Docker Compose orchestration. GitHub Actions CI with accuracy gate. Railway and Vercel deployment. Prometheus setup. | Full system deployed and publicly accessible. CI pipeline passing. | Low. |
| 20 | Evaluation | Full evaluation run. Classifier metrics. Fingerprint stability. Degradation validation. Failsafe tests. All results tables completed. All research questions answered. | Complete evaluation chapter results. Every RQ answered with data. | Low. |
| 21 | Writing | Dissertation: Introduction, Related Work, Design & Architecture, Implementation. | 4 chapters drafted. | Medium — writing takes longer than expected. Start earlier if possible. |

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| 22 | Writing | Dissertation: Evaluation, Discussion, Conclusion, Abstract. Proofread. Submit. | Final dissertation. Submission. | Low. |

# 14. Risk Register

| Risk | Likelihood | Impact | Contingency | Week |
|---|---|---|---|---|
| XGBoost macro-F1 below 0.78 on held-out circuits | Medium | High | First action: add frequency-domain features (FFT on throttle and speed traces). Second: reduce classification from 8 to 6 drivers (drop PER and SAI as most-confused midfield pair). Third: relax target to 0.72 and report with honest analysis. | 6 |
| FastF1 telemetry gaps for specific driver-circuit-season combinations | Low | Medium | Drop any combination with <30 clean laps after filtering. Minimum coverage enforced in Week 1. At most 2 combinations expected to drop. | 1 |
| X/Y position data unavailable for corner decomposition | Medium | Low | Corner decomposition skipped for affected sessions. Features F19–F21 and F25 set to NULL. Lap-level features F01–F18 still computed. Corner decomposition becomes an optional enhancement, not a requirement. | 9 |
| Three.js WebGL integration fails in deployment environment | Medium | Low | D3.js 2D fallback implemented and tested in Week 18. Fallback is invisible to user. Three.js is enhancement, not requirement. Style space still fully functional in 2D. | 18 |
| Degradation detector mean lead time below 2 laps | Medium | High | First: lower confirmation requirement from 2 consecutive laps to 1 lap. Second: reduce threshold to 1.5SD. Third: add F27 and F28 to deviation score earlier (reduce minimum stint from 8 to 6 laps). Report threshold sensitivity analysis regardless. | 12 |
| UMAP projection unstable for unknown drivers | Low | Medium | PCA fallback always maintained. If UMAP stability test fails (Procrustes variance > 0.05): switch to PCA as primary projection for unknown drivers. UMAP | 8 |

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| | | | retained for known driver style space where stability is acceptable. | |
| Personalisation engine recommendations too wide (all LOW confidence) | Medium | Medium | Report as finding: certain setup parameters (differential ramp, suspension stiffness) are not reliably inferrable from telemetry alone. Narrow scope to the 5 parameters with highest mapping confidence. This is an honest research contribution. | 13 |
| Supabase free tier storage limit reached | Low | Low | Store only feature aggregates and SHAP summaries in PostgreSQL. Raw telemetry Parquet files stored locally. Synthetic or intermediate data not persisted. | Ongoing |
| Dissertation writing time underestimated | High | High | Writing begins in Week 20 alongside evaluation. Evaluation chapter drafted in parallel with running evaluations. No new features after Week 19 regardless of temptation. | 20 |

# 15. Academic Contribution Statement

## 15.1 Novel Contributions

### Contribution 1 — Unified Driver-Car Compatibility Framework

APEX is the first publicly documented system to quantify driver-car compatibility as a continuous score derived from behavioural telemetry. Prior work treats driver identification and vehicle setup as separate problems. APEX unifies them through a shared fingerprint representation.

### Contribution 2 — Fingerprint Deviation as Tyre Degradation Leading Indicator

This work introduces the use of within-stint behavioural fingerprint deviation as a predictive signal for tyre degradation. The hypothesis — that degradation manifests in driving behaviour before lap time — is tested and validated against three seasons of official F1 race data. Results include lead time quantification, compound stratification, and threshold sensitivity analysis.

### Contribution 3 — Cross-Series Style Transfer via Continuous Style Space

The UMAP-based continuous style space allows drivers from feeder series or unknown sources to be projected into the same space as established F1 drivers without retraining. Unknown driver projection is a forward-pass operation on the trained manifold. This enables quantitative style comparison across car classes and team boundaries.

### Contribution 4 — Formal Failsafe Evaluation for Sports Telemetry ML

APEX formally characterises the failure modes of an ML-driven sports telemetry system and evaluates explicit mitigation mechanisms including confidence gating, circuit breakers, null tolerance policies, and fallback chains. This is the first published characterisation of this class of failures in the F1 domain. Research Question 5 is dedicated to this contribution.

## 15.2 Limitations (Explicitly Stated)

- Setup recommendations are approximated from telemetry, not measured from actual car setup data. This is stated in every recommendation output.
- The dataset covers 8 drivers and 10 circuits. Generalisability beyond this cohort is discussed but not empirically validated.
- The degradation detector is validated on historical replay, not live race conditions. Live performance may differ due to session-specific noise not present in clean historical data.
- UMAP projections are inherently stochastic. Stability is quantified and reported, but the projection is not deterministic across different random seeds without Procrustes alignment.

## 15.3 Publication Target

Contribution 2 (degradation detection) and Contribution 4 (failsafe evaluation) are individually suitable for submission to a sports analytics conference or workshop. Target venues: MIT Sloan

Sports Analytics Conference, StatsBomb Data Science Conference, or the KDD Sports Analytics Workshop.

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